

BOTOU, China

WMO: 546180

Lat: 38.083N

Lon: 116.550E

Elev: 13

StdP: 101.17

Time Zone: 8.00 (E08)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.5	-8.6	-24.8	0.4	-3.7	-22.5	0.5	-3.5	6.4	-0.3	5.6	0.1	1.3	290	0.411

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.1	35.4	23.3	34.1	23.5	32.8	23.3	27.7	31.8	26.8	30.8	26.0	29.8	3.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.6	22.1	30.3	25.7	20.9	29.5	24.8	19.9	28.6	88.8	31.7	84.5	31.1	80.8	29.9	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.5	5.5	4.8	-13.1	37.8	2.3	1.3	-14.8	38.8	-16.2	39.6	-17.5	40.3	-19.2	41.3		
(5)			-14.3	28.6	2.2	1.0	-15.9	29.3	-17.2	29.9	-18.5	30.4	-20.1	31.2	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.1	-2.1	1.6	8.5	15.3	21.4	26.0	27.7	26.2	22.0	15.3	6.6	-0.1	(6)
(7)		DBStd	10.97	3.30	3.88	4.53	4.02	3.39	3.05	2.37	2.35	3.05	3.76	4.29	3.27	(7)
(8)		HDD10.0	1131	374	235	82	5	0	0	0	0	0	4	118	312	(8)
(9)		HDD18.3	2559	632	468	304	107	11	1	0	0	7	106	352	571	(9)
(10)		CDD10.0	2632	0	0	37	164	353	480	548	502	362	170	16	0	(10)
(11)		CDD18.3	1018	0	0	1	17	106	231	289	243	118	13	0	0	(11)
(12)		CDH23.3	10307	0	0	10	148	982	2548	3318	2376	849	76	0	0	(12)
(13)		CDH26.7	4239	0	0	1	29	335	1214	1519	898	233	9	0	0	(13)
(14)	Wind	WSAvg	2.2	1.8	2.2	2.8	3.0	2.6	2.4	2.1	1.8	1.9	2.0	1.9	1.8	(14)
(15)	Precipitation	PrecAvg	544	3	8	10	24	36	79	172	125	40	30	15	3	(15)
(16)		PrecMax	750	24	35	41	62	75	207	337	288	100	185	60	14	(16)
(17)		PrecMin	279	0	0	0	0	4	18	28	24	4	0	0	0	(17)
(18)		PrecStd	132	5	9	11	19	20	41	78	64	27	34	16	4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.5	15.2	24.5	29.8	34.4	37.4	37.4	34.8	32.2	27.9	20.2	10.7	(19)
(20)			MCWB	2.6	6.9	13.4	17.6	19.7	21.2	23.6	27.4	22.8	18.2	12.0	4.1	(20)
(21)		2%	DB	6.6	12.5	21.6	27.0	31.9	35.6	35.5	33.1	30.7	25.4	17.4	8.3	(21)
(22)			MCWB	1.0	5.4	11.7	16.4	18.9	21.3	24.9	25.9	21.4	16.3	10.6	2.8	(22)
(23)		5%	DB	4.8	10.4	18.9	24.9	30.1	34.2	34.1	32.0	29.3	23.5	15.2	6.5	(23)
(24)			MCWB	0.0	4.2	10.3	15.1	18.3	21.1	25.0	25.1	20.8	15.3	9.4	1.9	(24)
(25)		10%	DB	3.2	8.2	16.3	22.9	28.4	32.7	32.7	30.9	27.9	21.6	13.1	4.9	(25)
(26)			MCWB	-0.9	2.9	8.5	13.9	17.7	20.9	24.8	24.6	20.0	14.4	8.2	1.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	8.5	15.7	20.4	22.5	25.3	28.7	29.0	25.7	20.1	13.3	5.7	(27)
(28)			MCD B	7.9	12.8	21.9	27.1	29.7	32.1	33.0	32.9	30.2	24.1	17.9	8.8	(28)
(29)		2%	WB	1.6	6.3	13.0	18.1	21.5	24.2	27.9	27.7	23.6	18.6	11.7	3.9	(29)
(30)			MCD B	5.6	10.8	19.7	24.7	28.9	30.9	32.2	31.4	28.0	22.4	15.6	6.7	(30)
(31)		5%	WB	0.5	4.8	11.1	16.6	20.4	23.4	27.0	26.8	22.5	17.2	10.3	2.7	(31)
(32)			MCD B	3.8	9.1	17.5	23.0	27.5	29.7	31.5	30.4	26.9	21.1	14.1	5.5	(32)
(33)		10%	WB	-0.4	3.5	9.1	15.1	19.3	22.6	26.3	26.0	21.5	15.9	9.0	1.6	(33)
(34)			MCD B	2.4	7.8	15.5	21.3	26.0	28.8	30.6	29.3	25.9	19.9	12.3	4.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.6	9.1	10.4	10.9	10.8	10.4	8.1	7.6	9.1	9.6	8.9	8.2	(35)
(36)			MCD BR	10.6	12.0	13.2	13.2	13.2	12.6	10.1	8.6	10.6	11.6	11.2	10.3	(36)
(37)			MCWBR	6.7	6.9	6.6	6.1	4.8	3.8	3.3	3.3	3.8	5.0	6.1	6.2	(37)
(38)		5% WB	MCD BR	9.2	10.8	12.4	12.0	11.1	10.1	7.9	7.5	8.8	9.2	9.8	8.7	(38)
(39)			MCWBR	6.2	6.8	6.8	6.8	5.3	3.9	3.6	3.6	4.1	4.9	6.0	5.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.502	0.574	0.586	0.632	0.671	0.810	0.796	0.762	0.689	0.668	0.562	0.505	(40)	
(41)		taud	1.778	1.665	1.653	1.604	1.574	1.394	1.460	1.502	1.576	1.571	1.715	1.808	(41)	
(42)		Ebn at Noon	614	624	677	679	664	576	580	583	591	539	550	573	(42)	
(43)		Edh at Noon	164	209	232	257	270	323	300	282	248	227	174	150	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.30	2.96	4.37	5.24	5.86	5.33	4.76	4.54	4.06	3.17	2.40	2.10	(44)	
(45)		RadStd	0.26	0.41	0.39	0.31	0.42	0.46	0.32	0.37	0.37	0.32	0.36	0.26	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (11 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page